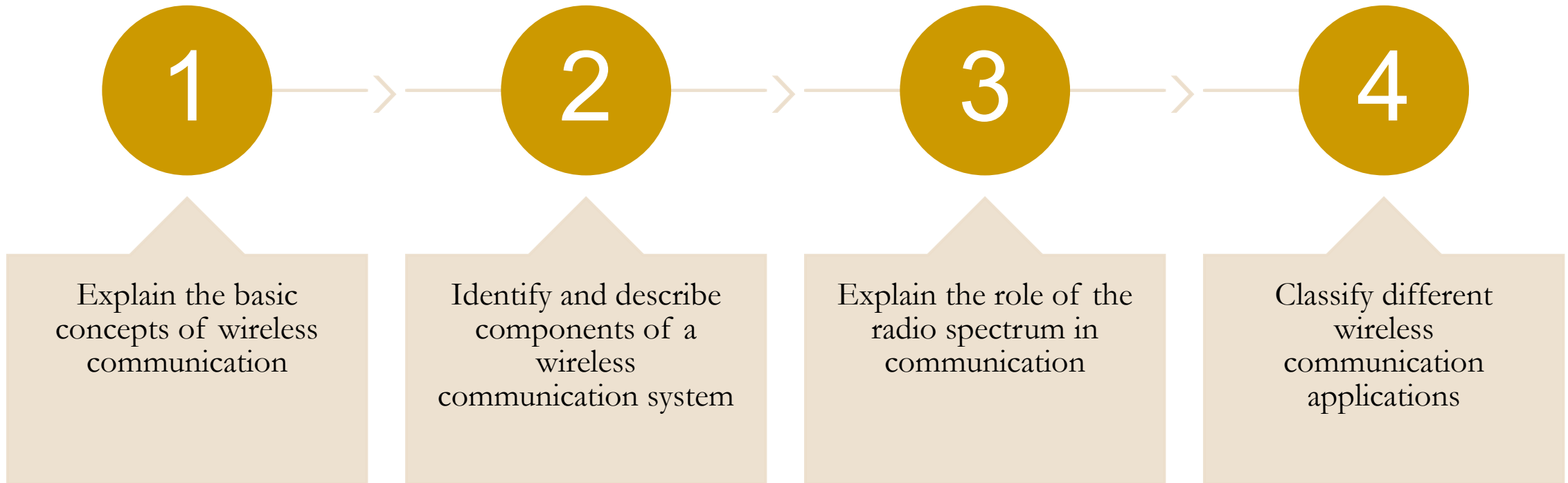


Chapter 1: Introduction

Learning Objectives

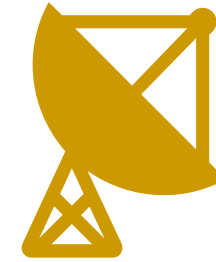


What is Wireless Communication?



Wireless communication is the **transfer of information** between two or more points **without physical connections**, using **electromagnetic waves**.

- The information from sender to receiver is carrier over a **well-defined** frequency band (**channel**)
- Each channel has a **fixed** frequency **bandwidth** and **Capacity** (bit-rate)
- Different channels can be used to transmit information in **parallel** and **independently**.



Assume a spectrum of 120 KHz is allocated over a base frequency for communication between stations A and B and each channel occupies 40 KHz

- We get 3 channels
- Each with B of 40 KHz

Station A	Channel 1 ($b - b+40$)	Station B
	Channel 2 ($b+40 - b+80$)	
	Channel 3 ($b+80 - b+120$)	

Why Wireless Communication?

Advantages

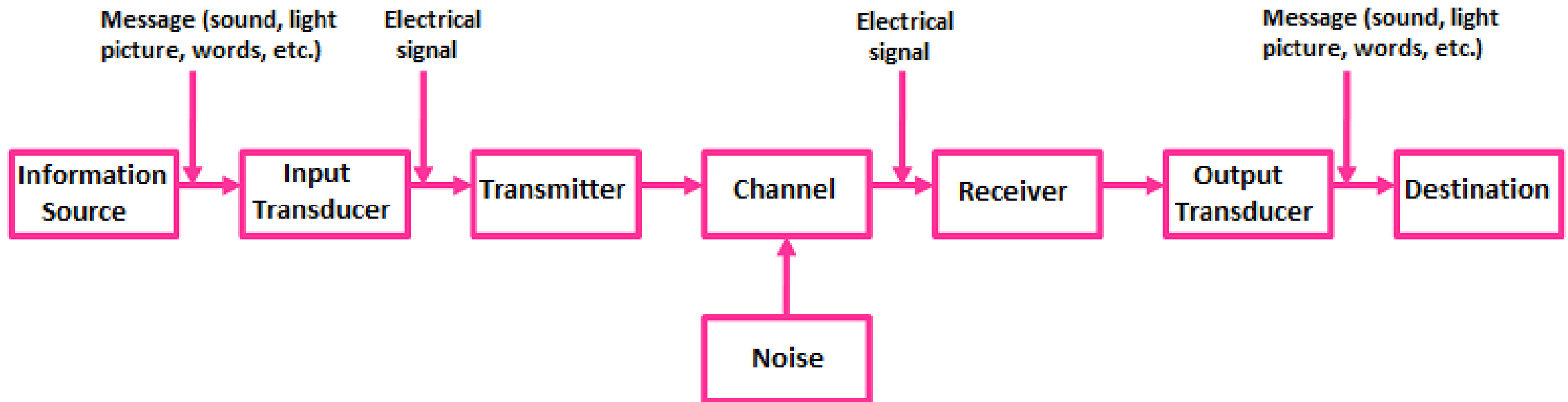
- Mobility and portability
- Easy deployment
- Cost-effectiveness
- Scalable networks
- Freedom from wires; Cost, Tidiness
- Flexibility e.g. Mobility

Limitations

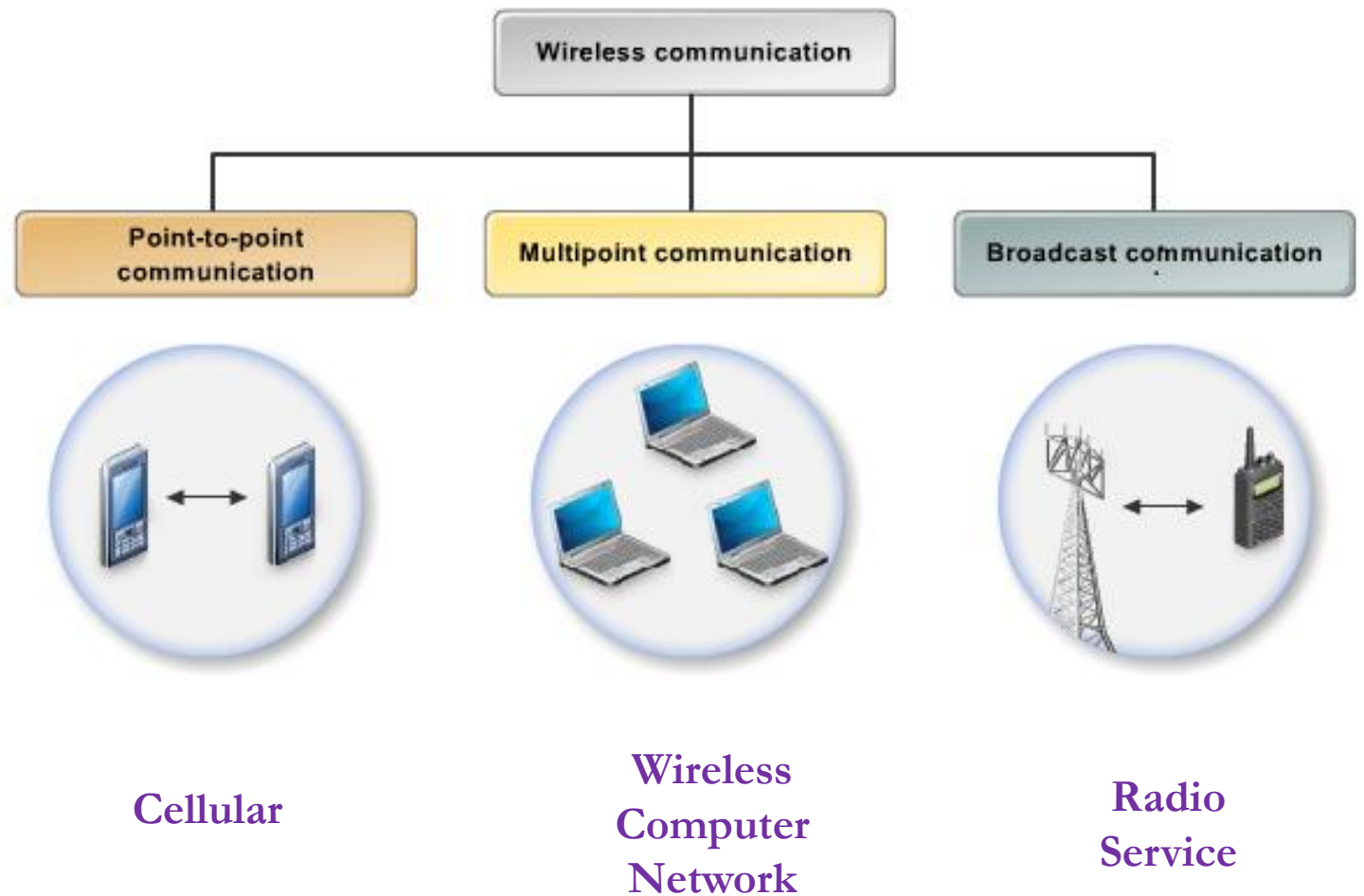
- Interference
- Limited bandwidth
- Security concerns
- Hand-off
- Location management



Basic Wireless Communication System



Types of Wireless Communication



Examples of Wireless Communications



Application	Range	Primary Use Case	Key Technology
Bluetooth	Short (~10m)	Connecting peripherals (headphones, mice).	Personal Area Network (WPAN)
Wi-Fi	Local (~100m)	High-speed internet for homes/offices.	Local Area Network (WLAN)
5G Cellular	Wide (km)	Mobile voice/high-speed data on the go.	Wide Area Network (WWAN)
Satellite	Global	Remote area coverage, GPS, global TV.	Non-Terrestrial Network (NTN)

Applications of Wireless Communication



Mobile Communication - Cellular networks (2G → 5G)

Broadcasting - Radio and TV transmission

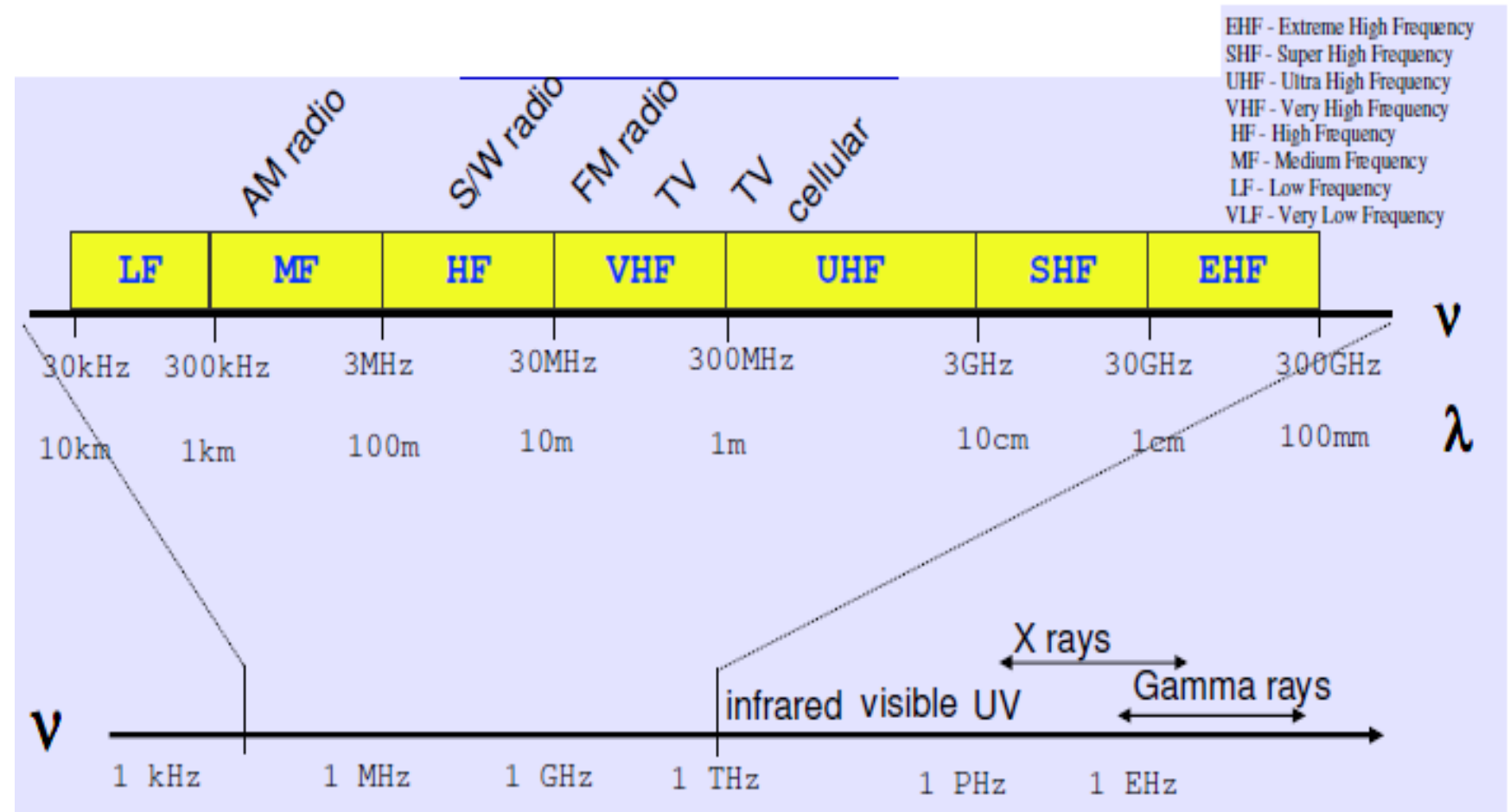
Satellite Communication - GPS, satellite TV, internet

Short-Range Communication e.g. WiFi, Bluetooth, Zigbee

Emerging Applications - Internet of Things (IoT), Smart homes, Smart cities

The Electromagnetic Spectrum

- The radio spectrum is the range of electromagnetic frequencies used for communication.
- Licensed vs. Unlicensed Bands: Some bands (like Wi-Fi and Bluetooth at 2.4 GHz) are "unlicensed," meaning anyone can use them following specific rules, leading to higher congestion but lower cost.



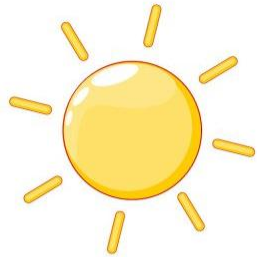
Cont...

Radioactive
sources



Medical x-rays

Sunlight



Visible light

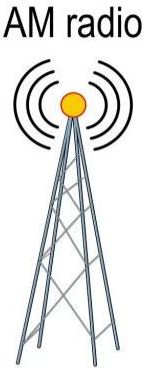
Thermal
infrared imaging



Microwave
oven



FM radio



AM radio

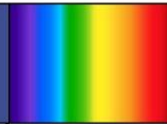
FREQUENCY



GAMMA -RAYS

X-RAYS

ULTRAVIOLET



INFRARED

MICROWAVE

RADIO WAVES

VISIBLE SPECTRUM

Wavelength
(nm)

380-450

450-480

480-510

510-575

575-585

585-620

620-780

Frequencies of Some Technologies

■ **Licensed bands** are commercial

- ✓ e.g GSM: 900 MHz, 1800 MHz
- ✓ 4G LTE allocations in Kenya

■ **Unlicensed bands** - Industrial, Scientific, and Medical (ISM)

- ✓ E.g. 2.4 GHz
 - IEEE 802.11b/g
 - Bluetooth
 - Zigbee
 - microwave oven
- ✓ 5.8 GHz ISM band:
 - IEEE 802.11a

Company	Frequency (MHz)	Bandwidth
Safaricom	800	20
Safaricom	1800	3
Telkom	800	20
Airtel	800	20
Faiba	700	28

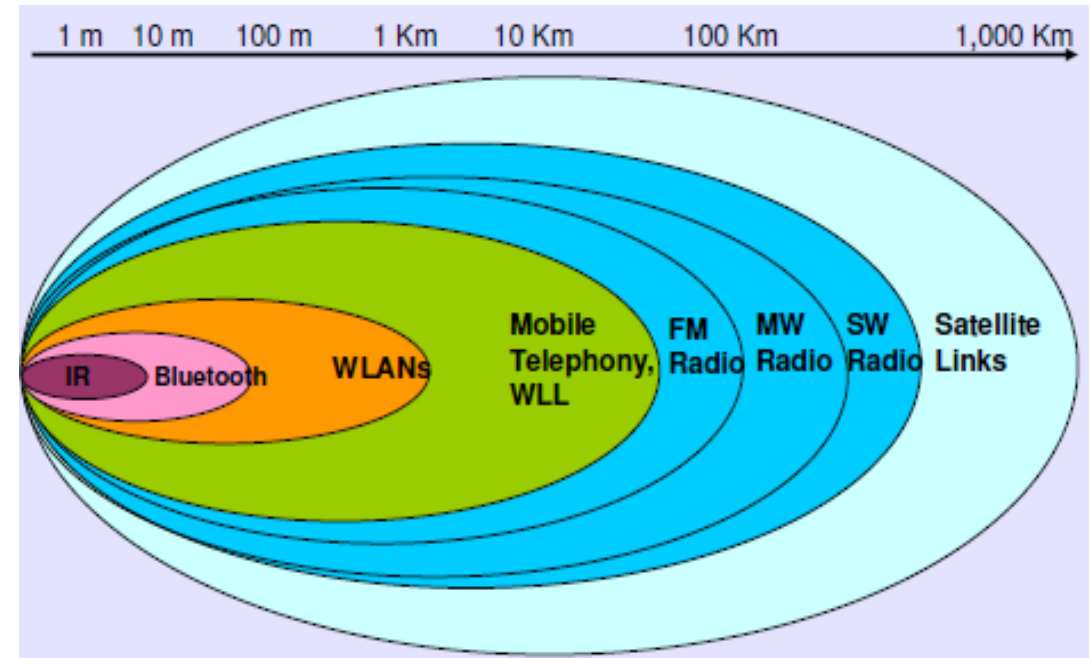
Types of Electromagnetic Carriers

- **Short distance** between the sender and receiver (e.g. TV box and a remote control)
 - ✓ **Infrared** waves
- **Long range distances** between sender and receiver (e.g. TV broadcasting and cellular service)
 - ✓ **Microwaves** and **Radio waves** are used
 - *Radio waves* are ideal when **large areas** need to be covered and **obstacles exist** in the transmission path
 - *Microwaves* are good when **large areas** need to be covered and **no obstacles** exist in the transmission path



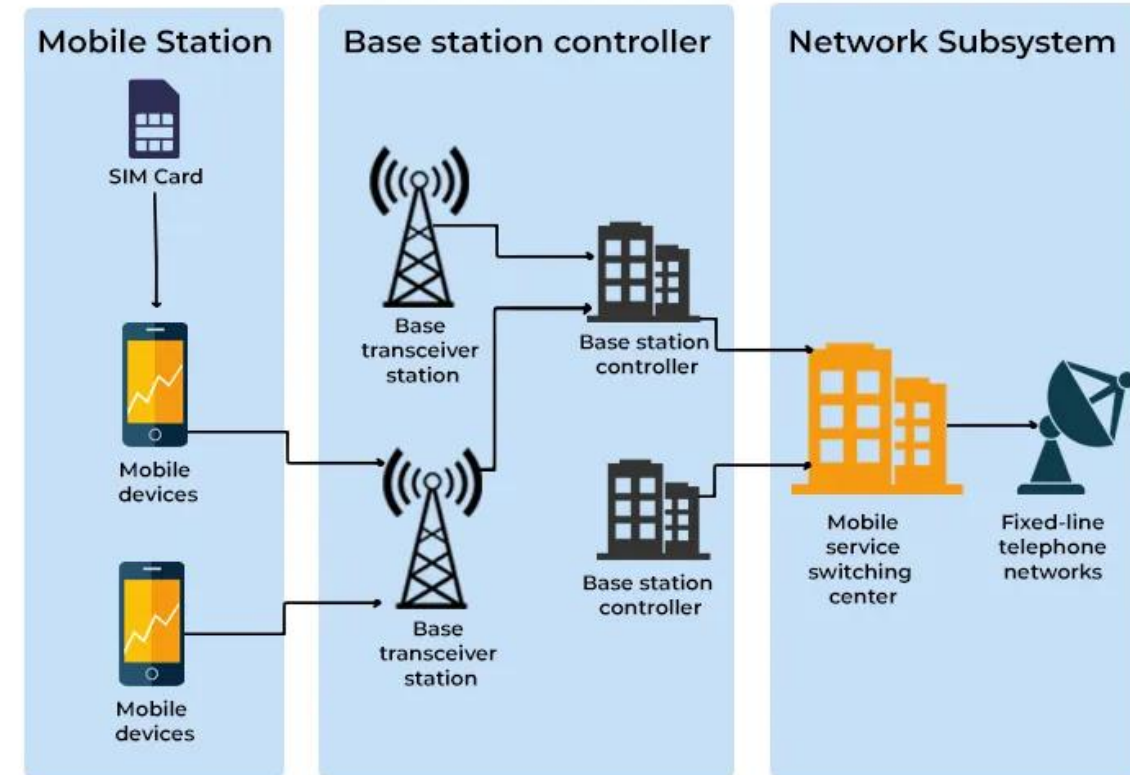
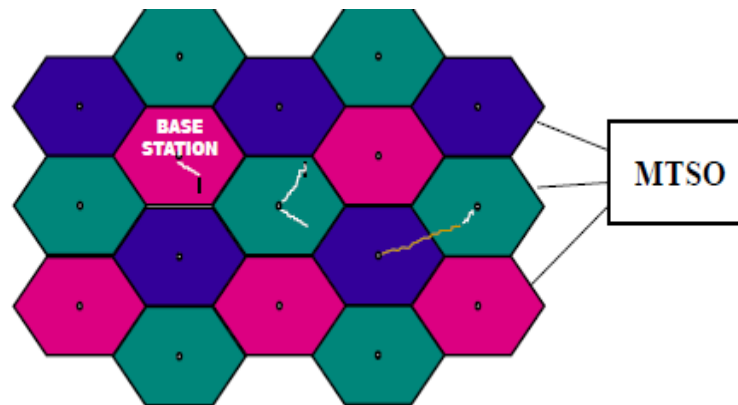
Current Wireless Systems

- Cellular systems
- Wireless LANs
- WiMAX
- Satellite systems
- Paging Systems
- Bluetooth
- Zigbee Radios, etc



Cellular Systems

- Reuse channels to maximize capacity
 - ✓ Geographic region divided into cells
 - ✓ Freq./timeslots/codes/space reused in different cells (reuse 1 common).
 - ✓ Interference between cells using same channel: interference mitigation key
 - ✓ Base stations/Mobile Telephone Switching Office (MTSOs) coordinate handoff and control functions
 - ✓ Shrinking cell size increases capacity, as well as complexity, handoff, ...



Trends in
Cell
Phones



Wireless LAN Standards (WiFi)

Multimedia Everywhere, Without Wires

- Streaming video, Gbps data rates, High reliability, Coverage inside and out

802.11b

- Standard for 2.4GHz ISM band (80 MHz)
- Frequency hopped spread spectrum
- 1.6-10 Mbps, 500 ft range

802.11a

- Standard for 5GHz NII band (300 MHz)
- OFDM with time division
- 20-70 Mbps, variable range (100-200ft)
- Similar to HiperLAN in Europe

802.11g

- Standard in 2.4 GHz and 5 GHz bands
- OFDM
- Speeds up to 54 Mbps and range of upto 200ft.

Bluetooth



Cable replacement RF technology (low cost)

Short range {10m (1mW), 100m (100 mW)}

Lower power than WiFi

2.4 GHz band (crowded)

1 Data (723.2 Kbps): Or 3 synchronous voice channels (up to 3Mbps)

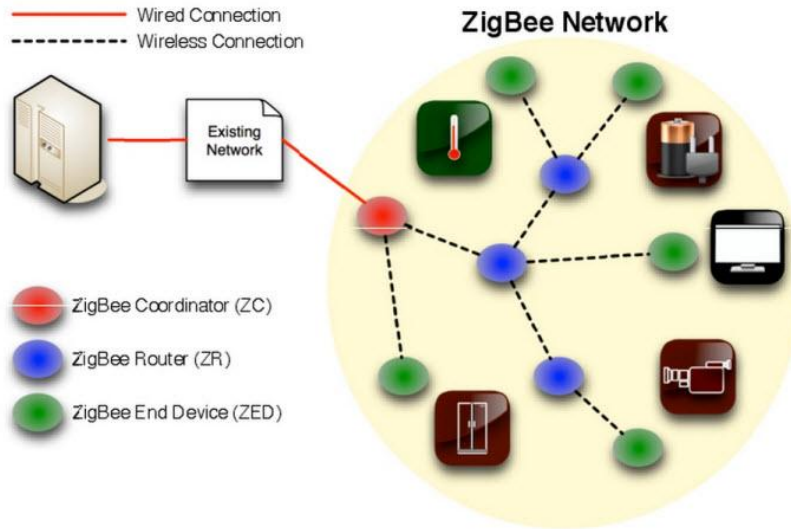
Frequency-hopping for multiple access with a carrier spacing of 1 MHz for 8 devices per pico-net.

Widely supported by telecommunications, PC, and consumer electronics companies.

Hands free phone (ear set) for cars, internet chat/VoIP

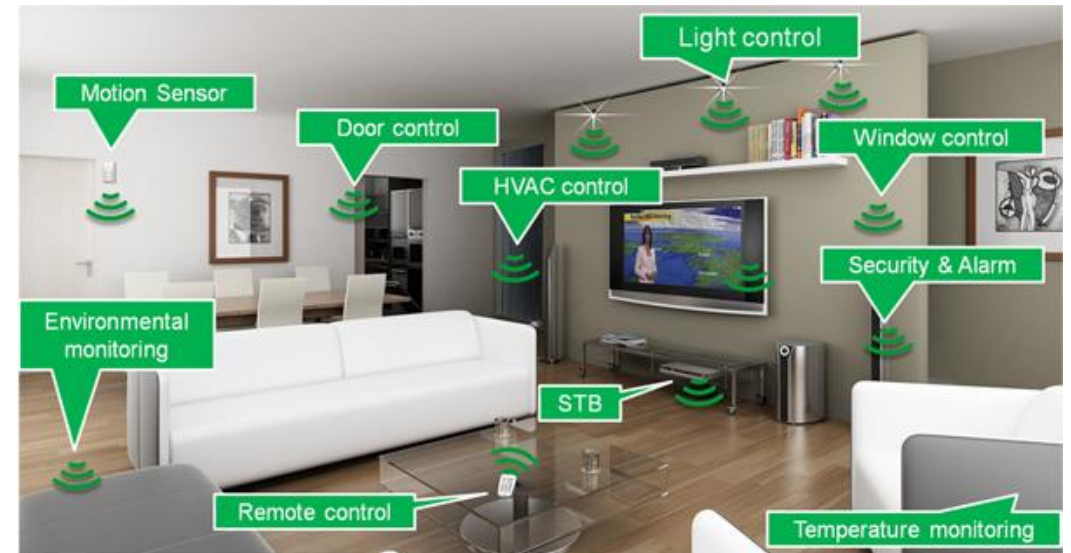
Intra-car networking announced by some car manufacturers in Europe.

ZigBee Radios

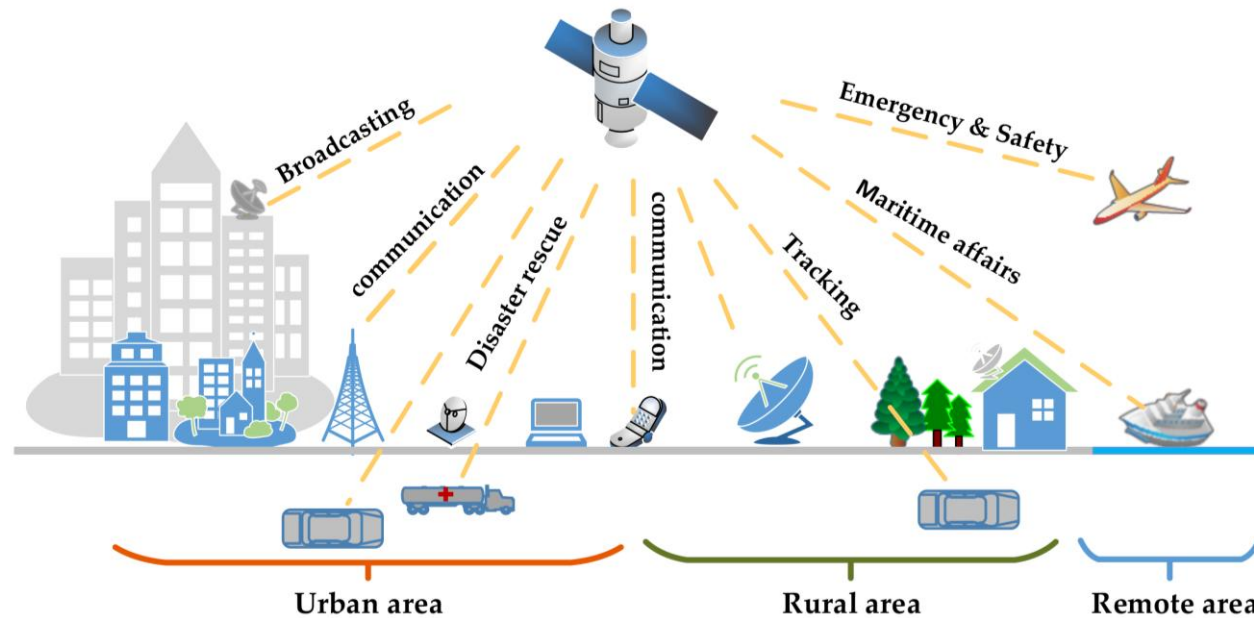


- Support for low latency devices
- CSMA-CA channel access
- Applications:
 - light switches, electricity meters, traffic management, and other low-power sensors.

- IEEE 802.15.4 Standard
- Low-rate low-power low-cost secure radio
 - ✓ Complementary to WiFi and Bluetooth
- Frequency bands: 784, 868, 915 MHz, 2.4 GHz
- Data rates: 20Kbps, 40Kbps, 250 Kbps
- Range: 10-100m line-of-sight
- Support for large mesh networking or star clusters



Satellite Systems



Cover very large areas

Different orbit heights

- GEOs (39000 Km) versus LEOs (2000 Km)

Optimized for one-way transmission

- Radio (XM, Sirius) and movie (SatTV, DVB/S) broadcasts
- Most two-way systems went bankrupt

Global Positioning System (GPS)
ubiquitous

- Satellite signals used to pinpoint location
- Popular in cell phones, PDAs, and navigation devices

Ad-Hoc Networks

Peer-to-peer communications

- No backbone infrastructure or centralized control

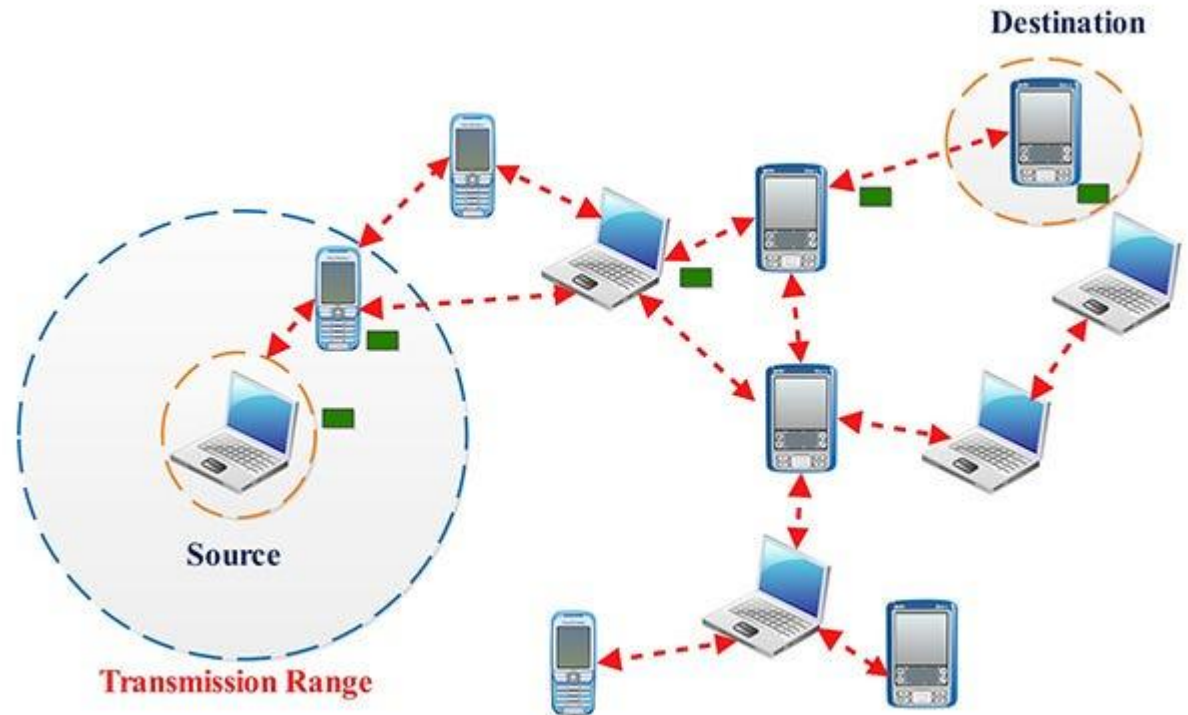
Routing can be multihop.

Topology is dynamic.

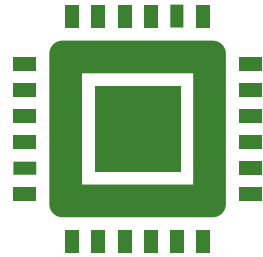
Fully connected with different link SNRs

Open questions/Challenges

- Fundamental capacity region
- Resource allocation (power, rate, spectrum, etc.)
- Routing



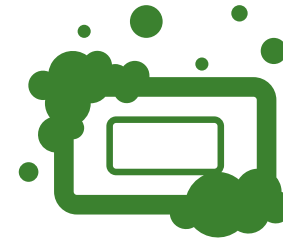
Cognitive Radios



Cognitive radios support new users in existing crowded spectrum without degrading licensed users

Utilize advanced communication and DSP techniques

Coupled with novel spectrum allocation policies



Multiple paradigms

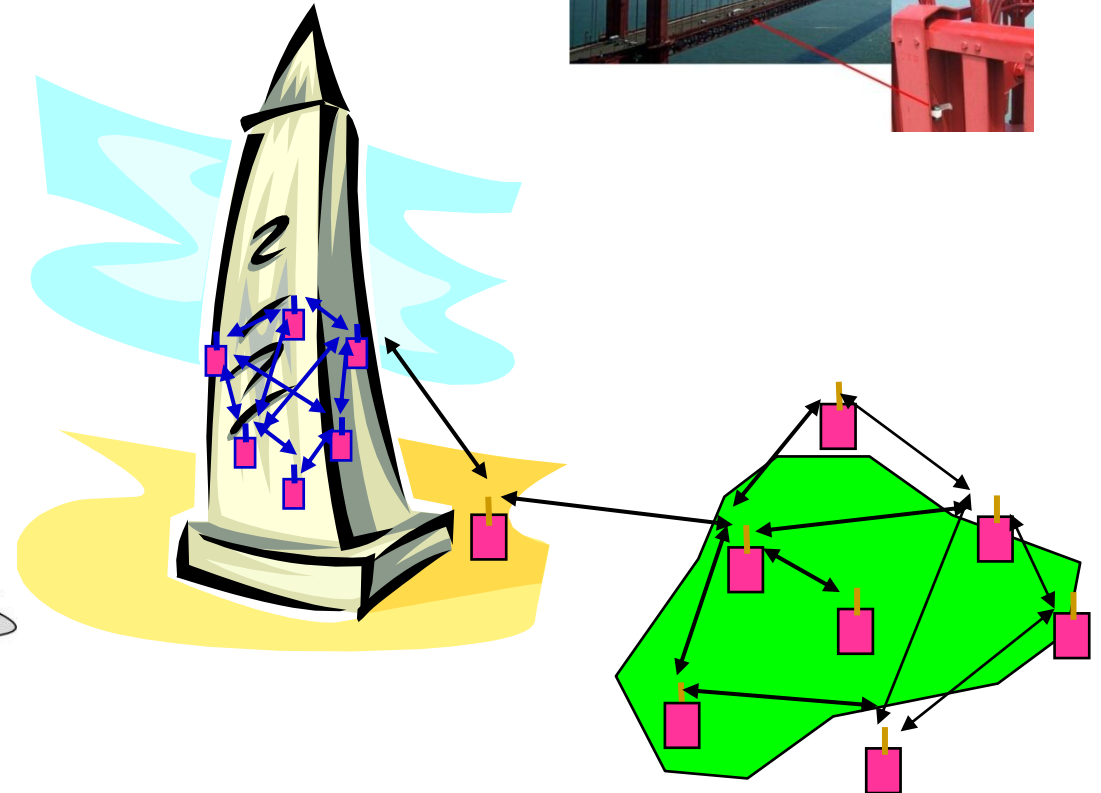
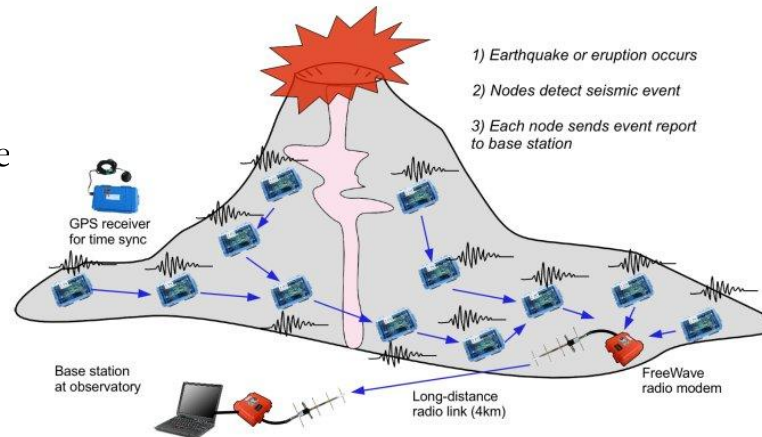
(MIMO) **Underlay** (interference below a threshold)

Interweave finds/uses unused time/freq/space slots

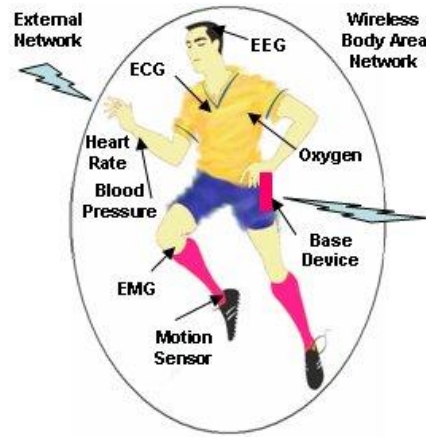
Overlay (overhears/relays primary message while cancelling interference it causes to cognitive receiver)

Wireless Sensor Networks

- Energy (transmit and processing) is the driving constraint
- Data flows to centralized location (joint compression)
- Low per-node rates but tens to thousands of nodes
- Intelligence is in the network rather than in the devices
- Used in:
 - ✓ Smart homes/buildings
 - ✓ Smart structures
 - ✓ Search and rescue
 - ✓ Homeland security
 - ✓ Event detection
 - ✓ Battlefield surveillance

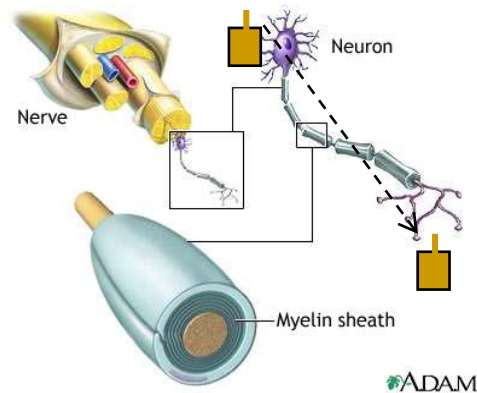


Applications in Health, Biomedicine & Neuroscience



Body-Area Networks

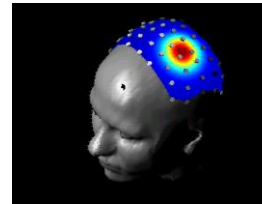
Recovery from Nerve Damage



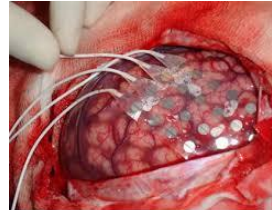
Neuroscience

- Nerve network (re)configuration
- EEG/ECoG signal processing
- Signal processing/control for deep brain stimulation
- SP/Comm applied to bioscience

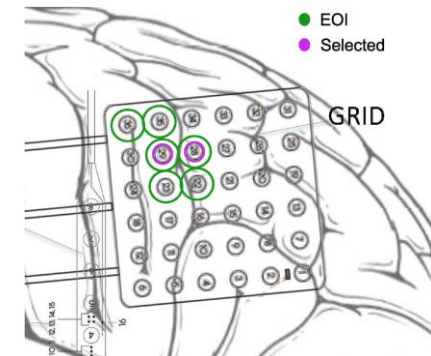
EEG



ECoG



ECoG Epileptic Seizure Localization



Trends in Application of Wireless Technologies

